

Surface-to>Returns

Visual Figure Package

Figures generated from curated reports, manifests, and model diagnostics.

SAMPLE

1996-2024

345 completed shards

SSVI GATES

100.0%

34,800 surfaces

CHARACTERISTICS

33

670,562 rows

SDF

0.116

227 OOS months

RETURN TEST

-0.15%

TAQ-net -0.37%/mo

Interpretation: option surfaces are a useful state representation; cost-aware return tests discipline the alpha evidence.

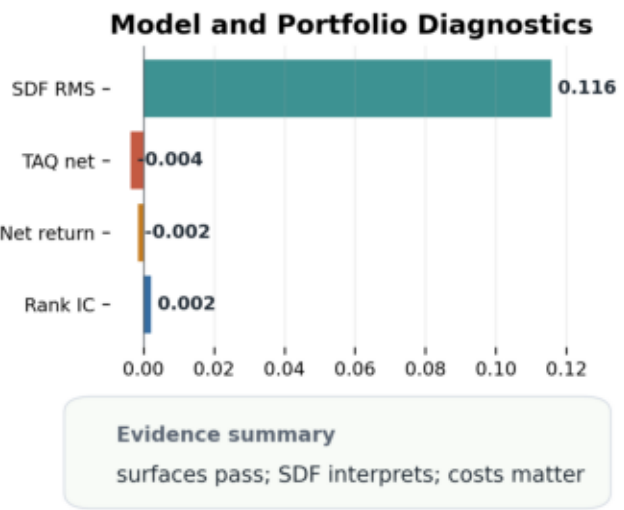
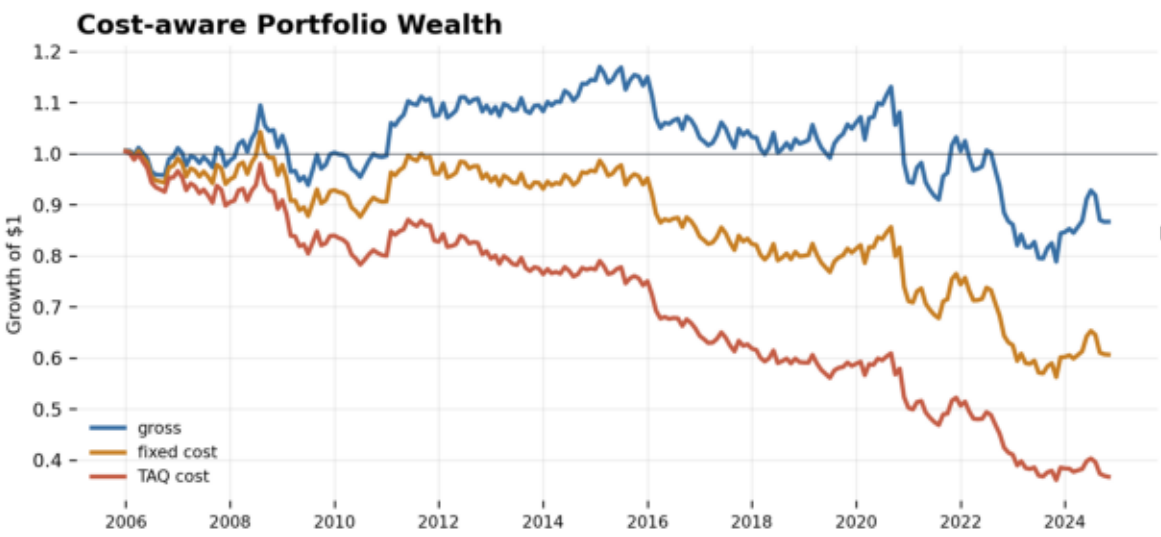
Visual Abstract

First-glance summary of the full-sample option-surface evidence package.

Surface-to>Returns

Full-sample option-surface asset-pricing evidence: construction, pricing kernel, mechanisms, and return tests

Finding: option surfaces are informative pricing states



Research Takeaway

- Option surfaces form a clean state representation.
- SSVI constraints hold across the full usable sample.
- SDF diagnostics and interpretation anchor the asset-pricing evidence.
- External state controls: FRED,BLS,BEA,EIA.
- Cost-aware portfolio tests discipline the return interpretation.

Empirical interpretation: option surfaces are valuable state variables for pricing-kernel and mechanism analysis.

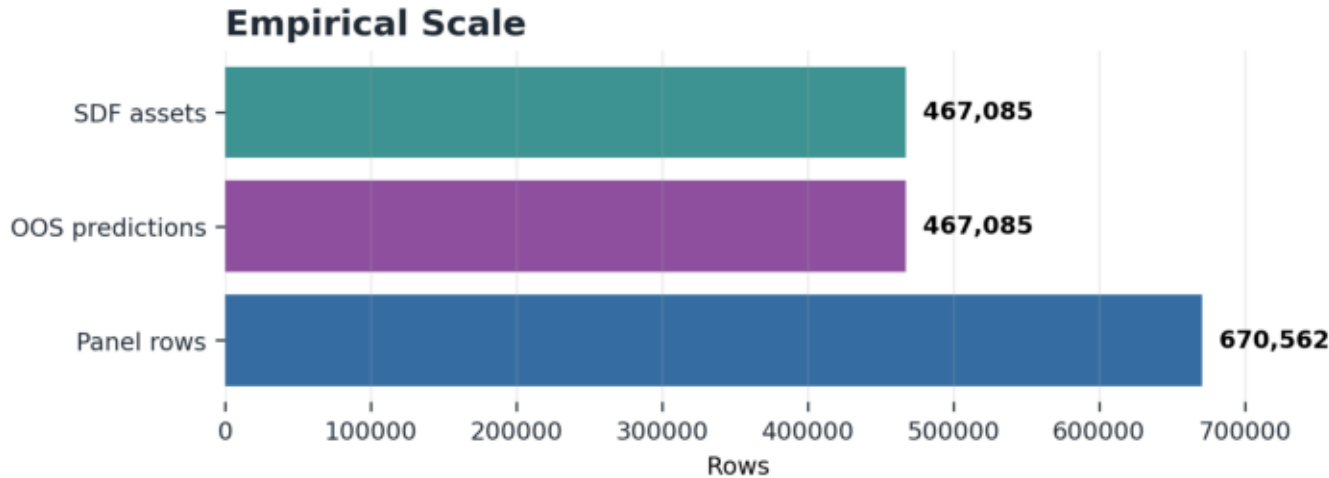
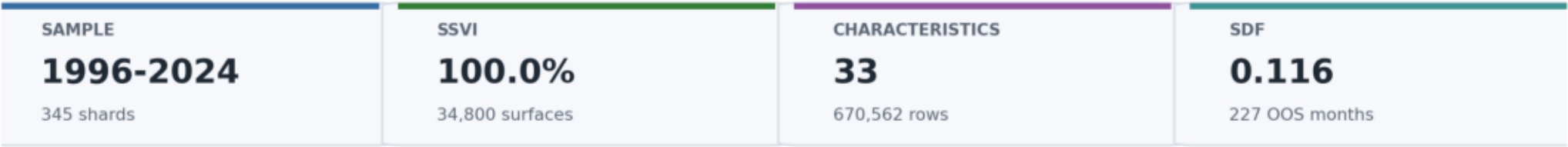
Cost-aware portfolio tests are reported alongside the pricing-kernel and mechanism evidence.

Figure 1. Data and Modeling Stack

WRDS extraction, linkage, surface construction, characteristics, costs, inference, and interpretation.

Figure 1. Data and Modeling Stack

WRDS extraction, no-arbitrage surface construction, characteristics, costs, inference, and interpretation at the usable 1996-2024 scale.

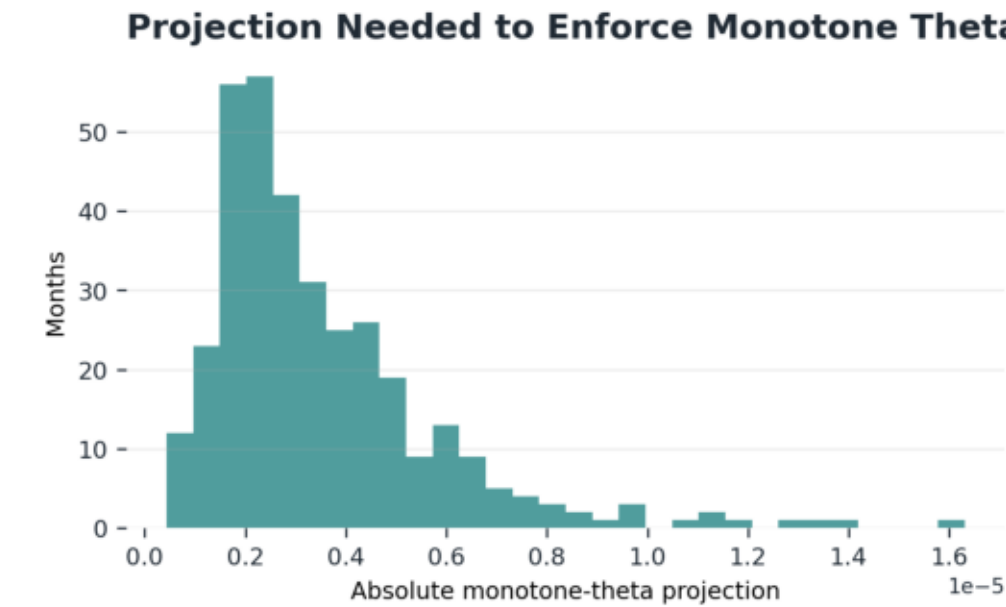
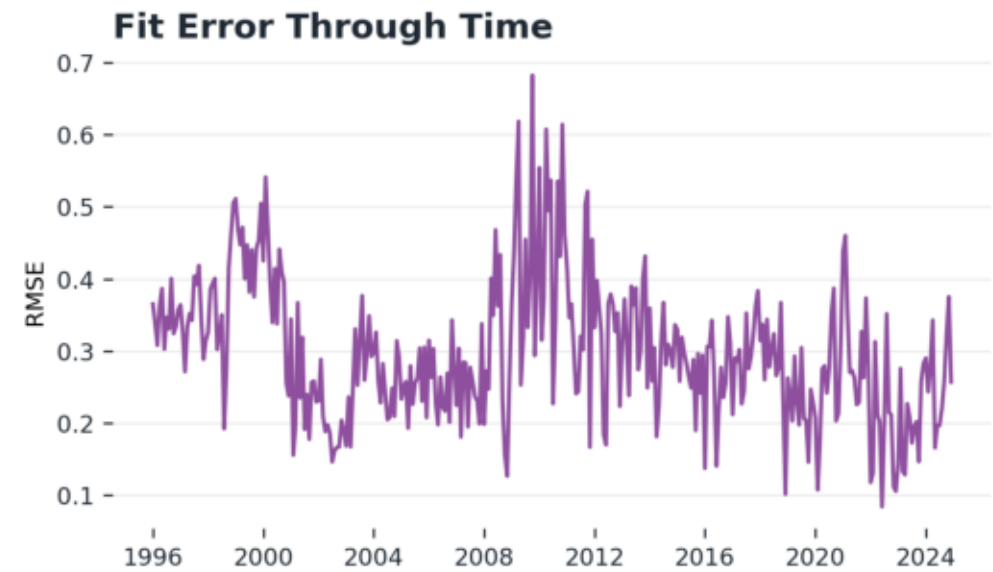
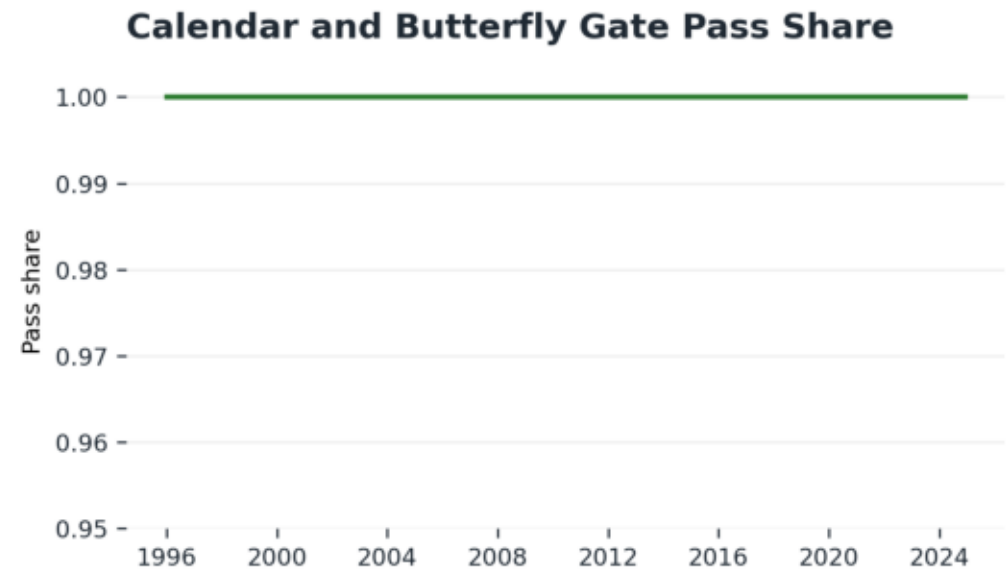


- Empirical Interpretation**
- The option-surface layer passes no-arbitrage gates at scale.
 - The SDF supplies interpretable pricing-kernel diagnostics.
 - Cost-aware portfolios place return tests in economic-cost context.

Figure 2. Surface Quality

SSVI pass share, fit error, monotone-theta projection, and no-arbitrage constraints.

Figure 2. SSVI Surface Quality and No-Arbitrage Diagnostics



Surface Evidence

- 34,800 of 34,800 date-security surfaces pass.
- Diagnostics enforce positivity, monotone theta, monotone theta-phi, calendar-grid monotonicity, and Gatheral-Jacquier bounds.
- Maximum theta projection in monthly report: 1.63×10^{-5} .

Figure 3. SDF and Interpretation

Three-branch conditional autoencoder SDF, integrated gradients, and TreeSHAP evidence.

Figure 3. Conditional Autoencoder SDF and Interpretation

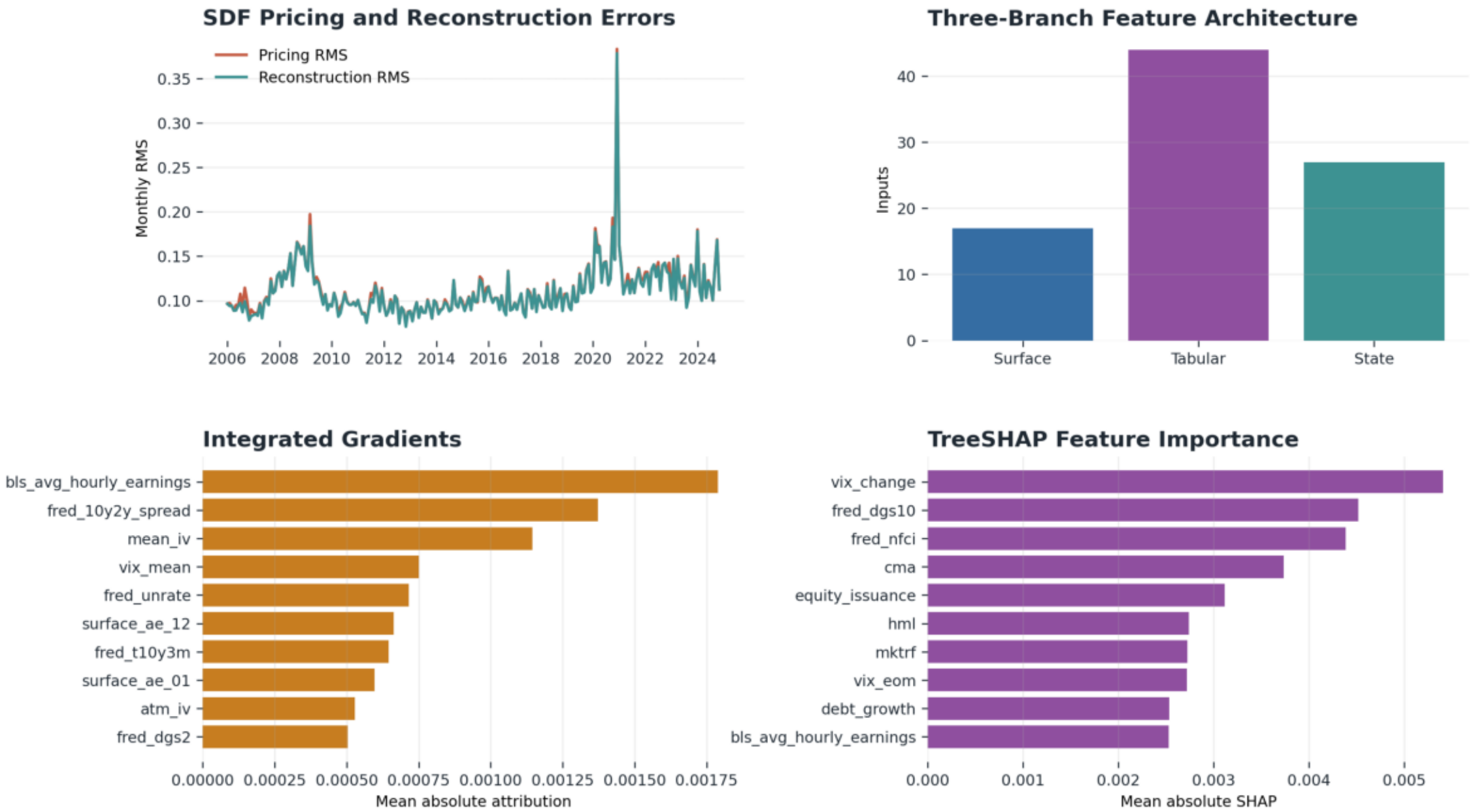


Figure 4. Return Evidence

Buffered long-short wealth, Sharpe ratios, factor alphas, and trading frictions.

Figure 4. Cost-Aware Return and Asset-Pricing Evidence

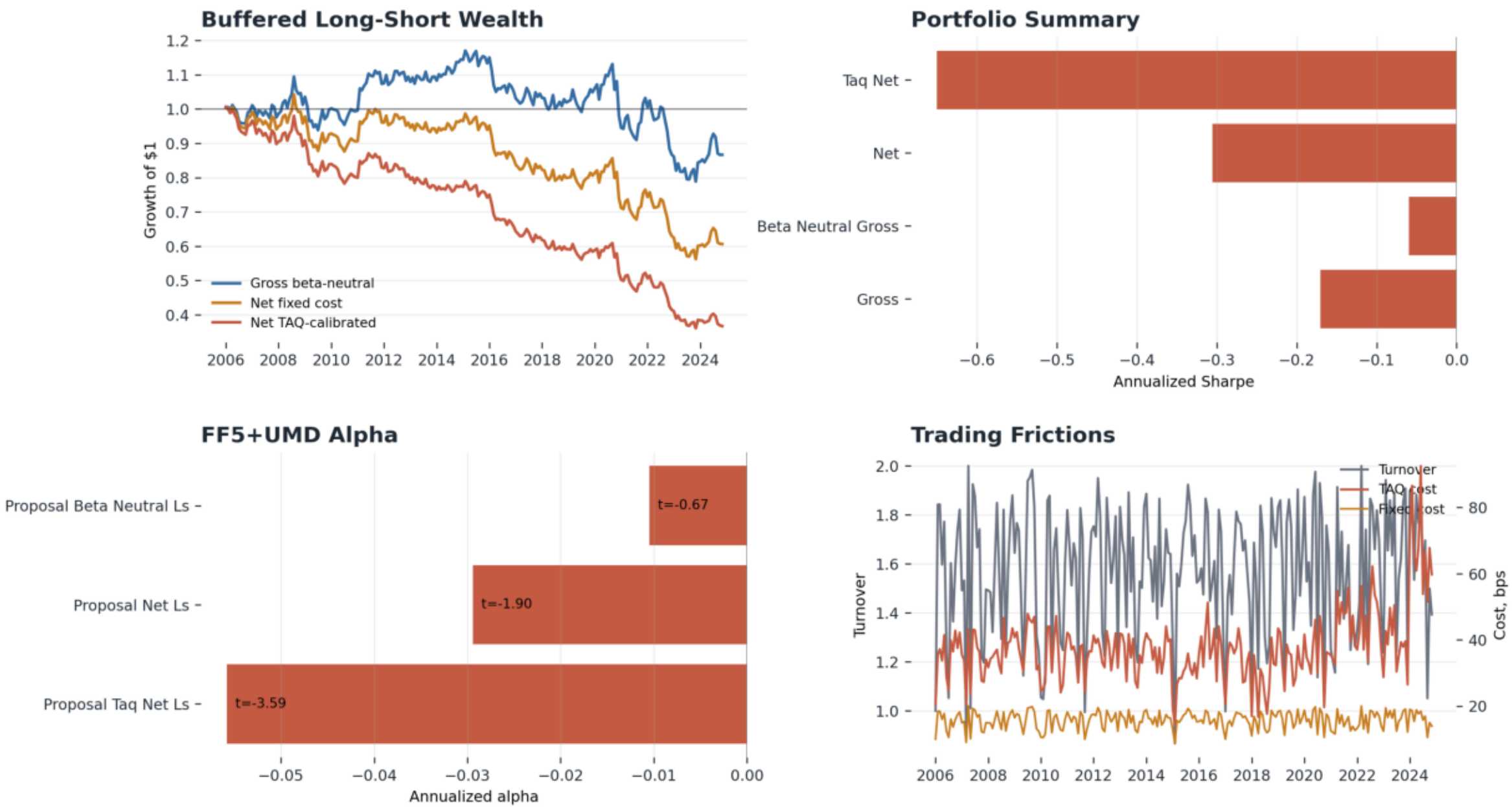
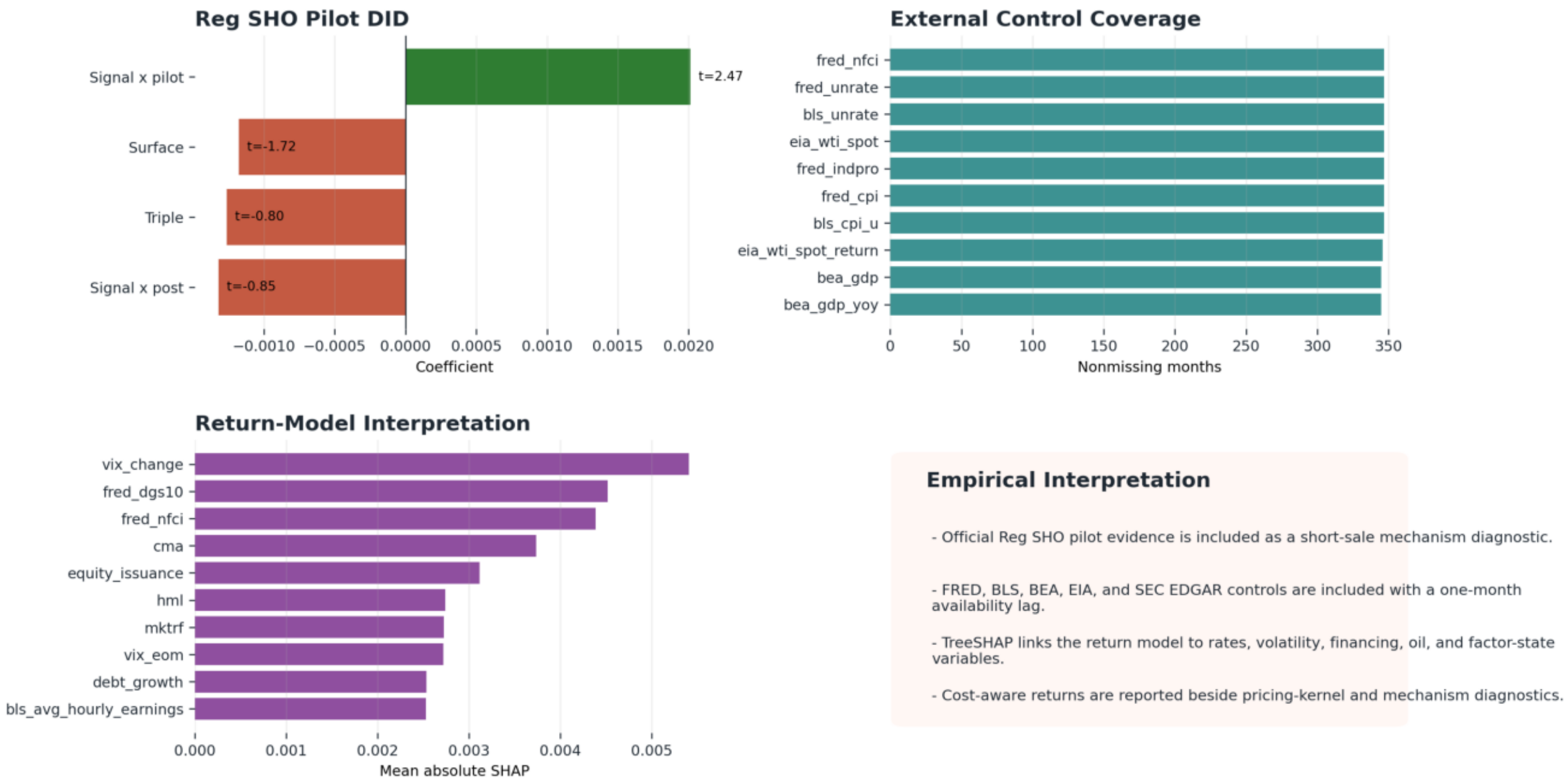


Figure 5. Mechanisms and Interpretation

Reg SHO mechanism evidence, external controls, SHAP interpretation, and empirical discipline.

Figure 5. Mechanisms, External Controls, and Interpretation



Proposal Evidence Pack

Detailed one-page view of diagnostics, model evidence, mechanisms, and interpretation.



Research Interpretation

The package connects option-surface states, pricing-kernel diagnostics, mechanism tests, and cost-aware returns.

Core Evidence

- Scaled WRDS-backed extraction and linkage through the usable 1996-2024 sample.
- SVI and SSVI no-arbitrage diagnostics pass at full usable scale.
- Characteristics, state controls, TAQ costs, IBES, Reg SHO, SDF, SHAP, and inference artifacts are long-short tests are reported as diagnostics beside the pricing-kernel evidence.
- Figures, report, deck, and PDF are generated from curated outputs and manifests.

Empirical Read

- Option surfaces work best here as state variables for pricing-kernel and mechanism analysis.
- TreeSHAP and integrated gradients connect the models to volatility, rates, financing, and factor-state variables.
- Cost-aware evidence.